



Explaining long-term bond yields synchronization dynamics in Europe[☆]

Jesús Crespo Cuaresma^{a,b,c,d,*}, Oscar Fernández^{a,e}

^a Vienna University of Economics and Business (WU), Department of Economics, Vienna, Austria

^b International Institute of Applied Systems Analysis (IIASA), Laxenburg, Austria

^c Austrian Institute of Economic Research (WIFO), Vienna, Austria

^d Wittgenstein Center for Demography and Global Human Capital (IIASA, VID/OEAW, WU), Vienna, Austria

^e Institute for Advanced Studies (IHS), Vienna, Austria

ARTICLE INFO

Dataset link: <https://data.mendeley.com/datasets/r3pwvjdfvh/1>

JEL classification:

C11

C33

C52

F45

H63

Keywords:

Long-term government bond yields

European monetary union

Synchronization measures

Bayesian model averaging

ABSTRACT

We present a novel empirical assessment of the determinants of sovereign yield synchronization dynamics in the European Monetary Union. This topic has been seldom addressed in the existing macroeconomic literature. We use a time-varying measure of government bond yields synchronization and Bayesian Model Averaging methods to show that the persistence of synchronization measures differs significantly between GIIPS countries (Greece, Ireland, Italy, Portugal and Spain) and the rest of the monetary union, as well as across periods characterized by whether the zero lower bound of interest rates was binding or not. The degree of synchronization in inflation rates with the rest of the currency area is a robust predictor of the synchronization of sovereign yields, as opposed to economic fundamentals describing the fiscal positions of individual countries. An out-of-sample forecasting exercise reveals that accounting for the most relevant economic fundamentals can lead to improvements in the directional accuracy of the forecasts of yield synchronization rates for GIIPS countries.

1. Introduction

The 2012 European sovereign debt crisis emerges as one of the most dramatic episodes in the recent economic history of the European Union. After the Great Recession, the dynamics of budget deficits in European economies ignited a lack of confidence in the ability of these countries to repay their debt. This was particularly true for the so-called GIIPS countries (Greece, Ireland, Italy, Portugal, and Spain), which experienced large increases in their long-term government bond yields and sovereign spreads, reflecting a loss of confidence of investors in these economies. Consequently, after the first phase of growing differentials in long-term interest rates that started in 2010, from 2012 the long-term interest rates of European countries diverged substantially. Such a development is depicted in Fig. 1, which shows the long-term government bond yield (10-year) spreads of GIIPS economies against the German bond for 1999–2019.

The objective of this paper is to unveil the robust determinants of synchronization dynamics in long-term government bond yields in

the European Monetary Union (EMU) over the last two decades in the presence of specification uncertainty. In particular, we take into account that the relationships between synchronization measures for sovereign bond yields and their determinants may be subject to parameter heterogeneity across individual economies and over time due to structural breaks and country-specific institutional settings. The driving factors of relative dynamics in government bond yields across European economies have not been rigorously addressed in the existing empirical literature hitherto, and the macroeconomic developments in the euro area over the last decade indicate that the lack of synchronization of government bond yields may hamper the sustainability of EMU as a currency area.

Related literature. The literature that examines the main empirical predictors of sovereign bond yields is extensive.¹ Kilponen et al. (2015) provide an excellent review of the literature devoted to studying the factors that determine sovereign yields and their change over time.

[☆] We would like to thank the Editor, the Associate Editor, and two anonymous referees for very helpful comments that significantly improved the paper. We would also like to thank the participants at the WU Economics Research Seminar and at the 16th International Conference in Computational and Financial Econometrics (CFE 2022) for insightful discussions on earlier versions of this paper.

* Correspondence to: Department of Economics, Vienna University of Economics and Business, Welthandelsplatz 1, 1020, Vienna, Austria.

E-mail address: jcrespo@wu.ac.at (J. Crespo Cuaresma).

¹ For theoretical contributions, see e.g. Arghyrou and Tsoukalas (2011) and Gapen et al. (2008) among others.

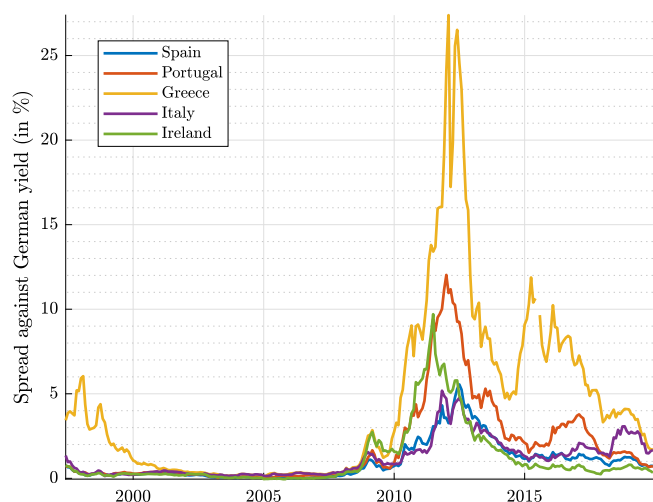


Fig. 1. Long-term government bond yield (10-year) spreads. The spread is computed as the difference between the yield of each country and the German yield.
Source: FRED dataset.

Variables describing the fiscal position of countries, such as the debt-to-GDP ratio, and the current account balance tend to be used as proxies for the creditworthiness and default risk of a country and are often identified as robust determinants of sovereign yields. Arghyrou and Kontonikas (2012), Aßmann and Boysen-Hogrefe (2011), Attinasi et al. (2009), Barrios et al. (2009) and Haugh et al. (2009) are important contributions that highlight the role of sovereign credit risk (as proxied by fiscal fundamentals) on government bond spreads in Europe. In an application using US data, Laubach (2009) finds that (projected) deficit and debt-to-GDP ratios are positively correlated with long-term interest rates. Poghosyan (2014) shows that increases in government debt-to-GDP ratios have positive impacts on government bond yields both in the long and short run. Real money market rates and inflation rates are also found to have significant effects on the deviation of yields from their long-run equilibrium. Poghosyan (2014) proposes two main mechanisms: the effect of higher debt levels on the default risk premium (see e.g. Manasse et al., 2003 for a theoretical underpinning of this mechanism) and the potential effect of fiscal expansions on the crowding out of private investment, which may imply higher real interest rates given the reduction in the capital stock (see e.g. the theoretical framework proposed by Engen and Hubbard, 2004).

Beyond the impact of macroeconomic fundamentals, much of the existing empirical literature emphasizes the role that contagion effects have on sovereign yields. Some of the countries in the European periphery (such as Portugal, Ireland, and Spain) have been found to suffer from economic spillovers from the crisis in Greece (Arghyrou and Tsoukalas, 2011; Claey's and Vasicek, 2012; Claey's and Vašíček, 2014). Previous work also argues that the effect of different sovereign risk factors on sovereign yields may be subject to time variation and regime changes (Kilponen et al., 2015), which justifies explicitly addressing parameter heterogeneity in econometric modeling exercises related to the assessment of the drivers of government bond yields. Arghyrou and Tsoukalas (2011) and Bernoth et al. (2012) find that the market penalizes adverse fiscal positions relatively more during crisis periods as compared to tranquil periods. In addition, Costantini et al. (2014) show that the effects of certain macroeconomic variables on the bond yield spread depend on whether the economies analyzed align with the standard optimum currency criteria, and report that gaps in competitiveness modulate the effect of fiscal variables on bond yield spreads.

Overall, these contributions identify specific sets of factors that might act as potential predictors of sovereign bond yield synchronization across countries. These can be broadly classified into four different

categories: (i) variables describing the fiscal position of countries, which act as indicators of their specific creditworthiness; (ii) covariates accounting for changes in the general aggregate sovereign risk; (iii) country-specific institutional characteristics that may affect the overall level of sovereign risk, and (iv) the business cycle characteristics of the particular economic period in consideration, since the effect of some factors may vary depending on whether the economy is in a recessionary or an expansionary phase.

With regard to the theoretical contributions in the literature, our paper is also related to the so-called “second-generation crisis models” (see e.g. Obstfeld, 1986, 1995). In these models, currency crises, defined as episodes in which the exchange rate depreciates substantially during a short period of time (Burnside et al., 2016), can occur due to the presence of self-fulfilling expectations that ignite speculative attacks on a particular economy (Burnside et al., 2016). The European Sovereign Debt (ESD) crisis might be understood through the lenses of these types of models given the self-fulfilling component of the financial market that played a non-negligible role in bringing certain Eurozone members into an adverse equilibria (De Grauwe and Ji, 2013) beyond what economic fundamentals would predict.

While the literature has mostly focused on examining the determinants of bond yield spreads, understanding the factors that drive bond yield synchronization levels within a monetary union emerges as a relevant question with important policy implications. If the cross-country synchronization of bond yield rates is weak within the union, supranational policy authorities may need to implement policies targeted only to those countries whose economies are detached from the other countries in the union. Moreover, to the extent that long-term government bond yields are a measure of how investors perceive sovereign risk in a given country, our exercise allows us to isolate the factors that robustly explain synchronization levels in sovereign risk as perceived by the financial sector. Hence our contribution lies in analyzing in depth a particular dimension of sovereign yields that has not been appropriately examined in the empirical literature so far.

Our second contribution to the literature relates to the methodological and modeling aspects of this paper. To the extent that the researcher is a priori unsure about which covariates matter when explaining the main variable of interest, we employ Bayesian Model Averaging (BMA) methods to account for specification uncertainty. This allows us to examine all potential models that relate our dependent variable of interest to a set of potentially relevant controls in a statistically rigorous and coherent manner.

Our results indicate that the level of the short-term interest rate shapes the persistence dynamics of synchronization measures of sovereign yields, with robust differences in periods dominated by the zero lower bound of interest rates and those periods following Mario Draghi's famous *whatever it takes* speech. This result also holds when comparing GIIPS countries to the rest of the monetary union. We find that economic fundamentals describing fiscal positions cannot systematically explain yield synchronization changes. Inflation synchronization dynamics appear robustly associated with the synchronization of long-term yields, and this association is, in turn, dependent on whether the economy is at the zero lower bound of interest rates, and differs for GIIPS countries as compared to the rest of EMU. We show that the associations uncovered making use of historical data lead to limited improvements in the directional accuracy of out-of-sample predictions of government bond yield synchronization measures.

The remainder of the paper is organized as follows. Section 2 describes the econometric framework used and gives an overview of the data employed. Section 3 shows the main results of the model averaging exercise. Section 4 reports the results of a pseudo-out-of-sample forecasting exercise aimed at evaluating the predictive accuracy of the models used, and Section 5 concludes.

2. Data and methodology

2.1. Bond yield synchronization in Europe

Our analysis assesses the empirical determinants of cross-country co-movement in government bond yields. For the purpose of measuring the level of synchronization in bond yields, we consider a time-varying measure based on sigma-convergence analysis (see Crespo-Cuaresma and Fernández-Amador, 2013a,b). The synchronization measure for country i (out of N countries that compose the monetary union)² in period t is given by

$$synch_{it} = \log(s_{it}) - \log(s_t), \quad (1)$$

where

$$s_t = \sqrt{\frac{\sum_j (r_{jt} - \bar{r}_t)^2}{N}}, \quad (2)$$

and

$$s_{it} = \sqrt{\frac{\sum_{j \neq i} (r_{jt} - \bar{r}_t)^2}{N-1}}. \quad (3)$$

The variable r_{it} denotes the long-term (10-year) government bond yield (in percentage points, sourced from the Federal Reserve Economic Dataset, FRED) of country i at time t . The variable s_{it} defined in Eq. (3) denotes the cross-country standard deviation in period t of the long-term government bond yield value (in percentage) excluding country i . In a similar way, the variable s_t defined in Eq. (2) denotes the standard deviation of long-term government bond yields including all the countries in the monetary union.³

The index builds upon the comparison of the variability in long-term bond yields across all countries of the monetary union with the variability in a counterfactual monetary union that does not include country i . If the index $synch_{it}$ is negative (positive), the cross-country yield variation is lower (higher) when excluding country i from the sample. In other words, negative (positive) values of this index imply that excluding country i results in a less (more) heterogeneous group of countries in terms of the dispersion of long-term government bond yields. This measure is relevant for our exercise because it allows us to construct country-specific, time-varying measures of synchronization that are more suitable than other measures found in the literature, since they allow us to take into account the (changing) structure of the monetary union in a more rigorous manner. We opt for our measure because our interest is not in yield synchronization of particular country pairs, but rather in the degree to which the bond yield volatility is higher or lower when a country is included in the union, relative to when it is not. Such an aim makes other existing measures based on correlations or pairwise comparisons (see those recently used by Hwang and Kim, 2021; Arčabić and Škrinjaric, 2021, , for instance) less intuitively appealing. In this sense, Barbieri et al. (2004) also emerges as a relevant contribution that analyzes bond yield synchronization in the Eurozone by employing principal component analysis (PCA). In contrast to Barbieri et al. (2004), we employ our measure because we are interested in the cross-country heterogeneity in bond yield synchronization, and because of its easier interpretability when compared to the standard PCA analysis. Other contributions in the literature, such as Martins (2022), examine bond yield synchronization among particular pairs of countries.

² Throughout all our empirical analysis (and due to data availability matters), the countries included are: Austria, Belgium, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Netherlands, Portugal, Slovakia, Slovenia, and Spain.

³ The variable r_t in Eq. (2) denotes the mean of the long-term yield at time t when all countries that form the union are included. Likewise, r_{it} in Eq. (3) denotes the mean of the long-term yield at time t excluding country i .

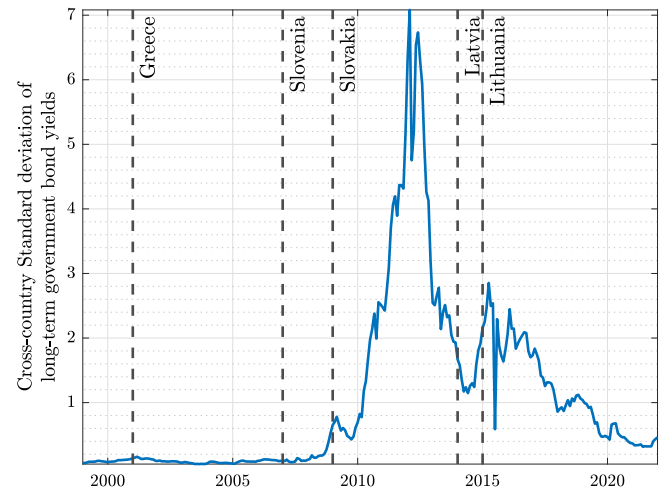


Fig. 2. Cross-country standard deviation of long-term government bond yields. Vertical lines indicate the date of accession to the EMU. Source: FRED dataset.

Fig. 2 depicts the cross-country standard deviation of the long-term government bond yields (i.e., the variable s_t in Eq. (2)) for the countries in EMU, along with the dates at which countries joined the currency area. The period with the highest levels of heterogeneity in bond yields corresponds to the government debt crisis that started in 2012. After this phase, the variation in the yields significantly decreases but does not return to the level observed prior to the crisis. The synchronization measure appears to present changes in its autocorrelation structure over the period considered, but is characterized by overall high persistence in the indicator during the last two decades.

Fig. 3 offers a visual representation of the synchronization variable (i.e., the variable $synch_{it}$ in Eq. (1)) for each country in the sample. As in the case of the overall synchronization measure presented in Fig. 2, the 2008 financial and the 2012 debt crises, as well as their aftermaths, are characterized by systematic decreases in yield synchronization for a large majority of the economies in the EMU. Certain countries (notably Greece and Italy) present negative values of the synchronization variable for a large number of periods. The decline in the synchronization variable after 2008 for Germany is related to German bonds being deemed a safe haven during this period, which led to a decrease in long-term yields in the country. Excluding Germany from the EMU would thus result in a less heterogeneous union, since many of the remaining countries experienced large increases in their long-term yields during this period, inducing higher levels of yield heterogeneity within the Union.

2.2. The determinants of bond yield synchronization: Assessing uncertainty

In order to assess empirically the robustness of the association between the covariates and the synchronization rate, we entertain regression models embedded in the specification given by

$$synch_{it} = \sum_{j=1}^4 \gamma_j synch_{i,t-j} + \sum_{k=1}^K \sum_{j=1}^4 \theta_{kj} x_{kit-j} + \alpha_i + \lambda_t + \epsilon_{it}, \quad \epsilon_{it} \sim N(0, \sigma^2), \quad (4)$$

where, in addition to the lagged dependent variable, other controls $\{x_{kt}\}$, $k = 1, \dots, K$ are used, potentially including lags of a macroeconomic uncertainty measure (Ahir et al., 2022), real GDP (in growth rates) to account for the overall business cycle dynamics, the current account balance of payments and debt over GDP to control for the fiscal position of the country with respect to the rest of the world, the annual rate of inflation (annual rate of change) to account for

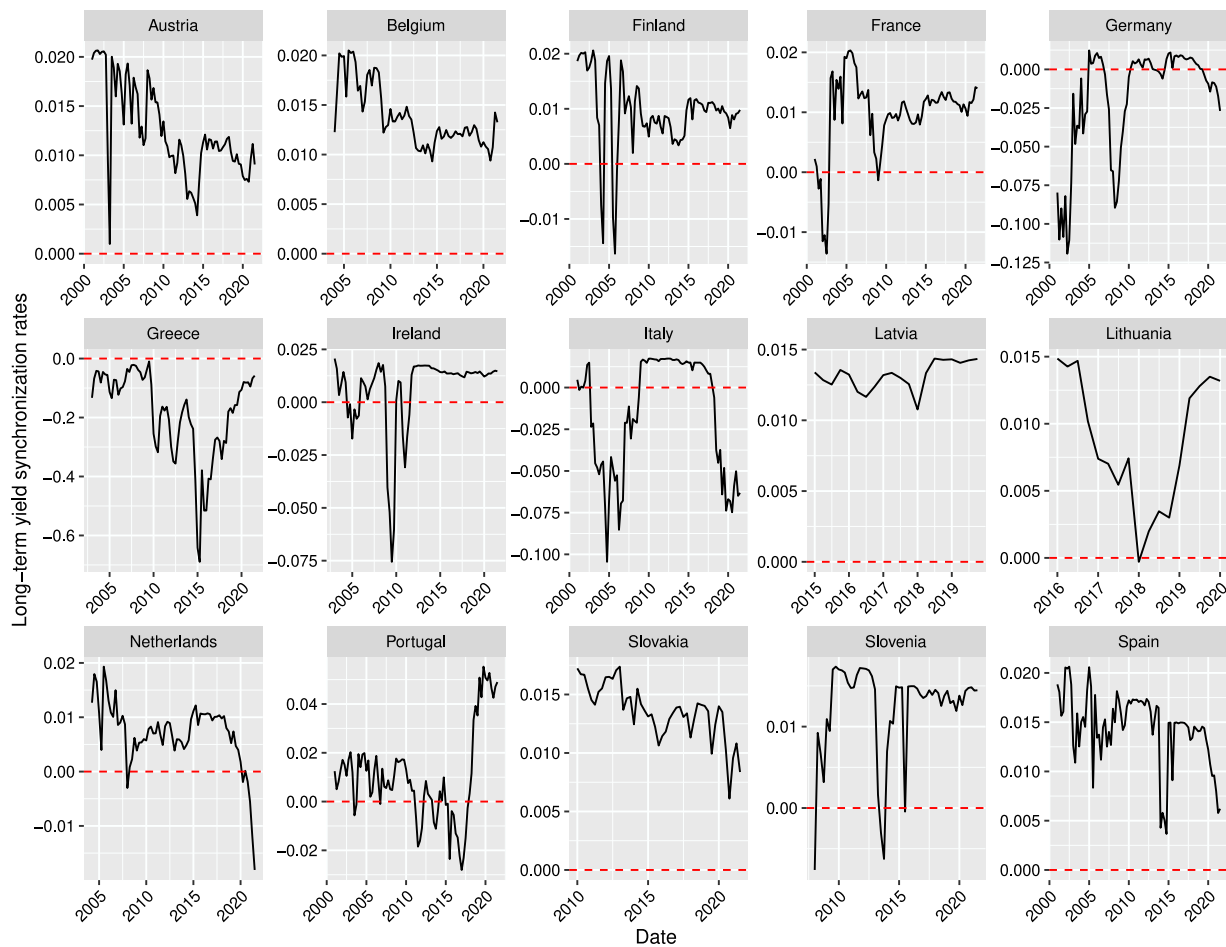


Fig. 3. Long-term government bond yield synchronization dynamics by country.

cross-country price differentials,⁴ as well as their interaction with (a) a dummy variable that identifies observations that belong to any of the GIIPS countries, (b) a (country-specific) dummy variable that identifies observations corresponding to recession periods,⁵ (c) a dummy variable that identifies observations for which the Euribor⁶ was below zero, so as to address the potential effects of zero lower bound effects on the determinants of bond yield synchronization, and (d) a dummy variable that identifies observations corresponding to those periods after the famous “whatever it takes” speech by Mario Draghi that took place on the 26th of July 2012. For a comprehensive description of the data, including summary statistics and the time range of the main variables for each country, see Tables A.1 and A.2 in the appendix.⁷

To the extent that we account for macroeconomic fundamentals and for a measure of economic uncertainty, our empirical exercise

allows us to examine both the effects of macroeconomic fundamentals versus the self-fulfilling component of the financial market on sovereign yield synchronization rates. Thus the addition of uncertainty measures is appropriate in this exercise because it potentially decreases the importance of macroeconomic fundamentals for the transmission of shocks into yield synchronization.

For all variables, up to four lags are allowed in the most general specification given that they are all sampled at a quarterly frequency. For our choice of covariates, this leads to a total of $K = 124$ potential (non-fixed effects) variables. Assuming that the country and time-fixed effects are always included in a model, the combination of those 124 variables leads to 2^{124} possible specifications nested in the general model in Eq. (4). The use of Bayesian Model Averaging (BMA) techniques allows us to account for specification uncertainty, thus evaluating the support that the data provides to the inclusion of particular variables as controls in the regression model in a rigorous and statistically founded manner.

The main purpose of the analysis is to compute the posterior distributions of an object of interest θ , which will typically be a parameter of the model while integrating away the uncertainty about specification choice,

$$P(\theta|y) = \sum_{f=1}^M P(\theta|y, M_f)P(M_f|y), \quad (5)$$

first justification of the use of dummy variables that take into account the corresponding structural breaks.

⁴ The data on real GDP, debt over GDP, and the annual inflation rate were obtained from Eurostat, while the data for the current account balance of payments were sourced from the OECD data.

⁵ Recession periods are sourced from the Federal Reserve Economic Data (FRED), which identifies these based on the “trough method”. We compute the values for Lithuania and Latvia making use of data from the OECD Composite Leading Indicators.

⁶ Data on the Euribor is sourced from the European Central Bank Statistical Data Warehouse.

⁷ For the sake of completeness, we depict in Table A.3 in the appendix the result of t-tests for the significance in the difference in means between recession and non-recession periods, ZLB and non-ZLB periods, and Draghi and non-Draghi periods, respectively, broken down by country. We show that for some countries the difference in means of the yield synchronization variable across periods is significantly different from zero, which provides a

Table 1
Bayesian model averaging results (covariates in levels).

	Time and country fixed effects			Standard BMA			Strong heredity prior		
	PIP	Post. Mean	Post. SD	PIP	Post. Mean	Post. SD	PIP	Post. Mean	Post. SD
$synch_{t-1}$	1	0,7941	0,084	1	0,9078	0,1123	1	1,1521	0,0464
$synch_{t-2}$	0,1281	-0,0284	0,0767	0,1814	-0,0436	0,0966	1	-0,3101	0,0596
$synch_{t-3}$	0,0029	-0,0002	0,0103	0,0159	-0,0019	0,027	0,2728	0,0438	0,0776
$synch_{t-4}$	0,0054	0,0006	0,0098	0,0685	0,0086	0,0367	0,7388	0,0908	0,0577
$uncert_{t-1}$	0,0008	0	0,0003	0,0009	0	0,0003	0,0005	0	0,0003
$uncert_{t-2}$	0,0008	0	0,0004	0,0013	0	0,0005	0,001	0	0,0004
$uncert_{t-3}$	0,0006	0	0,0003	0,0007	0	0,0003	0,0007	0	0,0003
$uncert_{t-4}$	0,0005	0	0,0002	0,0005	0	0,0002	0,0005	0	0,0002
gdp_{t-1}	0,0005	0	0,0003	0,0007	0	0,0003	0,0005	0	0,0002
gdp_{t-2}	0,0007	0	0,0003	0,0006	0	0,0003	0,0004	0	0,0002
gdp_{t-3}	0,0005	0	0,0002	0,0007	0	0,0003	0,0007	0	0,0003
gdp_{t-4}	0,0015	0	0,0013	0,0009	0	0,0007	0,0011	0	0,0006
bop_{t-1}	0,0007	0	0,0003	0,0008	0	0,0003	0,0006	0	0,0002
bop_{t-2}	0,0006	0	0,0003	0,0011	0	0,0004	0,0007	0	0,0003
bop_{t-3}	0,0006	0	0,0004	0,0008	0	0,0003	0,0007	0	0,0003
bop_{t-4}	0,0003	0	0,0002	0,0008	0	0,0003	0,0006	0	0,0003
$debt_{t-1}$	0,0005	0	0,0008	0,0131	-0,0004	0,0036	0,0067	-0,0002	0,0023
$debt_{t-2}$	0,0007	0	0,0014	0,0185	-0,0007	0,0074	0,0107	-0,0003	0,0039
$debt_{t-3}$	0,0004	0	0,0005	0,0118	-0,0003	0,0039	0,0059	-0,0001	0,0026
$debt_{t-4}$	0,0005	0	0,0012	0,0092	-0,0001	0,0057	0,0041	-0,0001	0,0021
inf_{t-1}	0,0004	0	0,0003	0,0005	0	0,0003	0,0004	0	0,0002
inf_{t-2}	0,0004	0	0,0003	0,0007	0	0,0003	0,0008	0	0,0004
inf_{t-3}	0,0004	0	0,0003	0,0008	0	0,0004	0,0006	0	0,0003
inf_{t-4}	0,0008	0	0,0006	0,0009	0	0,0006	0,001	0	0,0005
$I(rec_{it} = 1)$	0,0063	-0,0001	0,002	0,0033	0	0,0013	0,0082	-0,0001	0,0013
$I(rec_{it} = 1) \times synch_{t-1}$	0,97	0,2345	0,0834	0,6424	0,1226	0,1157	0,0059	0,0004	0,0057
$I(rec_{it} = 1) \times synch_{t-2}$	0,8282	-0,1438	0,0749	0,4606	-0,0863	0,1006	0,0003	0	0,0022
$I(rec_{it} = 1) \times synch_{t-3}$	0,0113	0,0013	0,0128	0,0221	0,0016	0,014	0	0	0,0003
$I(rec_{it} = 1) \times synch_{t-4}$	0,0541	0,0032	0,0142	0,4081	0,0239	0,0312	0,0003	0	0,0007
$I(rec_{it} = 1) \times uncert_{t-1}$	0,001	0	0,0004	0,0008	0	0,0004	0	0	0
$I(rec_{it} = 1) \times uncert_{t-2}$	0,0008	0	0,0003	0,0011	0	0,0005	0	0	0
$I(rec_{it} = 1) \times uncert_{t-3}$	0,0007	0	0,0003	0,0008	0	0,0005	0	0	0
$I(rec_{it} = 1) \times uncert_{t-4}$	0,0008	0	0,0003	0,0006	0	0,0003	0	0	0
$I(rec_{it} = 1) \times gdp_{t-1}$	0,0007	0	0,0003	0,0006	0	0,0002	0	0	0
$I(rec_{it} = 1) \times gdp_{t-2}$	0,001	0	0,0004	0,0008	0	0,0003	0	0	0
$I(rec_{it} = 1) \times gdp_{t-3}$	0,0009	0	0,0003	0,0011	0	0,0004	0	0	0
$I(rec_{it} = 1) \times gdp_{t-4}$	0,0344	0,0008	0,0047	0,0239	0,0006	0,0039	0	0	0
$I(rec_{it} = 1) \times bop_{t-1}$	0,0031	0,0001	0,0011	0,0036	0,0001	0,0011	0	0	0
$I(rec_{it} = 1) \times bop_{t-2}$	0,0159	0,0004	0,0032	0,0164	0,0004	0,003	0	0	0
$I(rec_{it} = 1) \times bop_{t-3}$	0,0032	0,0001	0,0011	0,0037	0,0001	0,0011	0	0	0
$I(rec_{it} = 1) \times bop_{t-4}$	0,0104	0,0002	0,0025	0,0116	0,0002	0,0024	0	0	0
$I(rec_{it} = 1) \times debt_{t-1}$	0,0111	-0,0003	0,0044	0,0265	-0,0008	0,0068	0	0	0
$I(rec_{it} = 1) \times debt_{t-2}$	0,0234	-0,0084	0,0841	0,0407	-0,005	0,0551	0	0	0
$I(rec_{it} = 1) \times debt_{t-3}$	0,0175	0,007	0,0796	0,0275	0,0025	0,0506	0	0	0
$I(rec_{it} = 1) \times debt_{t-4}$	0,0088	0,0004	0,0175	0,0226	-0,0002	0,0148	0	0	0
$I(rec_{it} = 1) \times inf_{t-1}$	0,0011	0	0,0005	0,0009	0	0,0006	0	0	0
$I(rec_{it} = 1) \times inf_{t-2}$	0,001	0	0,0009	0,001	0	0,0011	0	0	0
$I(rec_{it} = 1) \times inf_{t-3}$	0,0031	-0,0001	0,0015	0,0017	0	0,0013	0	0	0
$I(rec_{it} = 1) \times inf_{t-4}$	0,0042	-0,0001	0,0017	0,0031	-0,0001	0,0014	0	0	0
$I(GIIPS_i = 1)$	-	-	-	0,0039	-0,0001	0,0014	0,0013	0	0,0005
$I(GIIPS_i = 1) \times synch_{t-1}$	0,0345	0,011	0,0604	0,3455	0,1204	0,1713	0	0	0,0005
$I(GIIPS_i = 1) \times synch_{t-2}$	0,0433	-0,0144	0,0737	0,3719	-0,1548	0,2112	0	0	0
$I(GIIPS_i = 1) \times synch_{t-3}$	0,0182	0,0033	0,0254	0,3313	0,0602	0,1011	0	0	0
$I(GIIPS_i = 1) \times synch_{t-4}$	0,0023	0,0002	0,005	0,143	0,0207	0,059	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-1}$	0,0008	0	0,0005	0,0015	0	0,0008	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-2}$	0,0005	0	0,0003	0,0009	0	0,0006	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-3}$	0,0008	0	0,0004	0,0018	0	0,0007	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-4}$	0,0004	0	0,0002	0,0008	0	0,0004	0	0	0
$I(GIIPS_i = 1) \times gdp_{t-1}$	0,0011	0	0,0004	0,0008	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times gdp_{t-2}$	0,0007	0	0,0003	0,0006	0	0,0002	0	0	0
$I(GIIPS_i = 1) \times gdp_{t-3}$	0,0008	0	0,0003	0,0012	0	0,0004	0	0	0
$I(GIIPS_i = 1) \times gdp_{t-4}$	0,0036	0,0001	0,0011	0,0023	0	0,0008	0	0	0
$I(GIIPS_i = 1) \times bop_{t-1}$	0,0005	0	0,0002	0,0014	0	0,0005	0	0	0
$I(GIIPS_i = 1) \times bop_{t-2}$	0,0006	0	0,0003	0,0014	0	0,0005	0	0	0
$I(GIIPS_i = 1) \times bop_{t-3}$	0,0007	0	0,0004	0,0007	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times bop_{t-4}$	0,0006	0	0,0003	0,001	0	0,0004	0	0	0
$I(GIIPS_i = 1) \times debt_{t-1}$	0,0005	0	0,0009	0,0333	-0,0009	0,0075	0	0	0
$I(GIIPS_i = 1) \times debt_{t-2}$	0,001	-0,0002	0,0101	0,046	-0,0045	0,0447	0	0	0
$I(GIIPS_i = 1) \times debt_{t-3}$	0,0007	0	0,0031	0,0305	-0,0007	0,0168	0	0	0
$I(GIIPS_i = 1) \times debt_{t-4}$	0,0007	0,0001	0,0094	0,0281	0,0025	0,0412	0	0	0
$I(GIIPS_i = 1) \times inf_{t-1}$	0,0007	0	0,0005	0,0006	0	0,0003	0	0	0

(continued on next page)

Table 1 (continued).

$I(GIIPS_i = 1) \times inf_{i-2}$	0,0013	0	0,0008	0,0006	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times inf_{i-3}$	0,0006	0	0,0004	0,001	0	0,0005	0	0	0
$I(GIIPS_i = 1) \times inf_{i-4}$	0,0016	0	0,001	0,0007	0	0,0003	0	0	0
$I(ZLB_i = 1)$	0,0006	0	0,001	0,0006	0	0,0011	0,9568	0,0093	0,0124
$I(ZLB_i = 1) \times synch_{i-1}$	1	-1,0803	0,0722	1	-1,0906	0,0728	1	-1,0936	0,07
$I(ZLB_i = 1) \times synch_{i-2}$	1	0,9435	0,1183	1	0,977	0,089	1	1,0111	0,0747
$I(ZLB_i = 1) \times synch_{i-3}$	0,1944	0,0354	0,0735	0,0568	0,0088	0,0384	0,0002	0	0,003
$I(ZLB_i = 1) \times synch_{i-4}$	0,003	-0,0003	0,0053	0,0201	-0,0024	0,0177	0,0031	-0,0002	0,0051
$I(ZLB_i = 1) \times uncert_{i-1}$	0,0008	0	0,0006	0,0007	0	0,0004	0	0	0
$I(ZLB_i = 1) \times uncert_{i-2}$	0,0006	0	0,0004	0,0009	0	0,0004	0	0	0
$I(ZLB_i = 1) \times uncert_{i-3}$	0,0005	0	0,0004	0,0007	0	0,0005	0	0	0
$I(ZLB_i = 1) \times uncert_{i-4}$	0,0007	0	0,0004	0,0007	0	0,0004	0	0	0
$I(ZLB_i = 1) \times gdp_{i-1}$	0,0006	0	0,0002	0,0004	0	0,0002	0	0	0
$I(ZLB_i = 1) \times gdp_{i-2}$	0,0013	0	0,0005	0,0013	0	0,0005	0	0	0
$I(ZLB_i = 1) \times gdp_{i-3}$	0,0006	0	0,0002	0,0005	0	0,0002	0	0	0
$I(ZLB_i = 1) \times gdp_{i-4}$	0,0006	0	0,0004	0,0005	0	0,0002	0	0	0
$I(ZLB_i = 1) \times bop_{i-1}$	0,0006	0	0,0003	0,0006	0	0,0002	0	0	0
$I(ZLB_i = 1) \times bop_{i-2}$	0,0006	0	0,0002	0,0007	0	0,0003	0	0	0
$I(ZLB_i = 1) \times bop_{i-3}$	0,0007	0	0,0003	0,0009	0	0,0003	0	0	0
$I(ZLB_i = 1) \times bop_{i-4}$	0,0006	0	0,0003	0,0008	0	0,0003	0	0	0
$I(ZLB_i = 1) \times debt_{i-1}$	0,0012	0	0,0014	0,0011	0	0,0013	0	0	0,0003
$I(ZLB_i = 1) \times debt_{i-2}$	0,0012	0	0,004	0,001	0	0,0019	0	0	0,0005
$I(ZLB_i = 1) \times debt_{i-3}$	0,0017	0,0001	0,0081	0,0011	0,0001	0,0098	0	0	0,0002
$I(ZLB_i = 1) \times debt_{i-4}$	0,0013	0	0,0058	0,0012	-0,0001	0,0094	0	0	0,0001
$I(ZLB_i = 1) \times inf_{i-1}$	0,0007	0	0,0005	0,0007	0	0,0004	0	0	0
$I(ZLB_i = 1) \times inf_{i-2}$	0,0024	-0,0001	0,0014	0,0009	0	0,0006	0	0	0
$I(ZLB_i = 1) \times inf_{i-3}$	0,0075	-0,0003	0,004	0,0032	-0,0001	0,0026	0	0	0
$I(ZLB_i = 1) \times inf_{i-4}$	0,0007	0	0,001	0,0009	0	0,0007	0	0	0
$I(Draghi_i = 1)$	0,0011	0	0,0017	0,0009	0	0,0017	1	-0,0039	0,0126
$I(Draghi_i = 1) \times synch_{i-1}$	1	0,8569	0,0852	1	0,7924	0,1011	1	0,8196	0,0813
$I(Draghi_i = 1) \times synch_{i-2}$	1	-0,8641	0,1432	0,9999	-0,8032	0,1617	1	-0,7654	0,0876
$I(Draghi_i = 1) \times synch_{i-3}$	0,7775	0,1549	0,0885	0,5808	0,1189	0,1127	0,0065	0,0021	0,0279
$I(Draghi_i = 1) \times synch_{i-4}$	0,006	-0,0008	0,0124	0,0841	-0,017	0,0603	0,0065	-0,0013	0,0184
$I(Draghi_i = 1) \times uncert_{i-1}$	0,0021	0	0,0011	0,0012	0	0,0007	0	0	0
$I(Draghi_i = 1) \times uncert_{i-2}$	0,0015	0	0,0008	0,0013	0	0,0007	0	0	0
$I(Draghi_i = 1) \times uncert_{i-3}$	0,0006	0	0,0004	0,0006	0	0,0003	0	0	0
$I(Draghi_i = 1) \times uncert_{i-4}$	0,0005	0	0,0003	0,0006	0	0,0003	0	0	0
$I(Draghi_i = 1) \times gdp_{i-1}$	0,0004	0	0,0002	0,0006	0	0,0002	0	0	0
$I(Draghi_i = 1) \times gdp_{i-2}$	0,0008	0	0,0004	0,0009	0	0,0005	0	0	0
$I(Draghi_i = 1) \times gdp_{i-3}$	0,0005	0	0,0002	0,0006	0	0,0002	0	0	0
$I(Draghi_i = 1) \times gdp_{i-4}$	0,0006	0	0,0011	0,0007	0	0,0006	0	0	0,0004
$I(Draghi_i = 1) \times bop_{i-1}$	0,0007	0	0,0003	0,0006	0	0,0002	0	0	0
$I(Draghi_i = 1) \times bop_{i-2}$	0,0005	0	0,0002	0,0007	0	0,0003	0	0	0
$I(Draghi_i = 1) \times bop_{i-3}$	0,0009	0	0,0004	0,001	0	0,0004	0	0	0
$I(Draghi_i = 1) \times bop_{i-4}$	0,0008	0	0,0003	0,0009	0	0,0003	0	0	0
$I(Draghi_i = 1) \times debt_{i-1}$	0,0011	0	0,0034	0,0011	0	0,0013	0	0	0
$I(Draghi_i = 1) \times debt_{i-2}$	0,001	0	0,0035	0,001	0	0,0019	0	0	0,0001
$I(Draghi_i = 1) \times debt_{i-3}$	0,0014	0	0,0017	0,0013	0	0,0014	0	0	0,0001
$I(Draghi_i = 1) \times debt_{i-4}$	0,0011	0	0,0017	0,001	0	0,002	0	0	0,0002
$I(Draghi_i = 1) \times inf_{i-1}$	0,0004	0	0,0003	0,001	0	0,0005	0	0	0
$I(Draghi_i = 1) \times inf_{i-2}$	0,0005	0	0,0004	0,0011	0	0,0007	0	0	0
$I(Draghi_i = 1) \times inf_{i-3}$	0,0076	-0,0007	0,009	0,0114	-0,001	0,0099	0	0	0
$I(Draghi_i = 1) \times inf_{i-4}$	0,0129	0,0011	0,0108	0,0316	0,0021	0,0139	0	0	0,0001

Note: The table depicts the results for specifications where the yield synchronization variable is regressed against the *synchronization levels* of the independent variables. PIP stands for Posterior Inclusion Probability, Post. Mean is the mean of the posterior distribution of the corresponding effect, and Post SD is the standard deviation of the posterior distribution.

where $P(\Theta|y)$ denotes the posterior distribution of Θ , and $P(\Theta|y, M_f)$ denotes the posterior distribution of Θ conditional on the model specification M_f (out of all specifications $M = 2^{124}$). $P(M_f|y)$ denotes the posterior model probability, which is proportional to the product of the marginal likelihood, $P(y|M_f)$ and the prior probability of specification M_f ,

$$P(M_f|y) \propto P(y|M_f)P(M_f). \quad (6)$$

Collecting all the parameters associated with the explanatory variables in the specification given in equation Eq. (4) into the vector ψ_f and after the choice of priors for this vector, the marginal likelihood of the model can be obtained as

$$P(y|M_f) = \int_{\psi_f} \int_{\sigma} l(y|\psi_f, \sigma, M_f) P(\psi_f, \sigma) d\psi_f d\sigma, \quad (7)$$

where $l(y|\psi_f, \sigma)$ denotes the likelihood function of model M_f and $P(\psi_f, \sigma)$ is the prior density over the parameters of the specification.

Following the standard choice of priors over model-specific parameters employed in the BMA literature (see for example Fernández et al., 2001a,b), we impose an uninformative prior on the relative scale (that is, on $\ln \sigma$), and the *g*-prior by Zellner (1986) on the parameters associated to the covariates of the model,

$$p(\sigma) \propto \frac{1}{\sigma}, \quad (8)$$

$$p(\psi_f|\sigma) \sim \mathcal{N}\left(0, g\sigma^2(X_f^T X_f)^{-1}\right), \quad (9)$$

where X_f is the matrix of observations of the explanatory variables in the specification, and g is a scalar that needs to be elicited.

For our empirical application, we employ three different setups. In the first space of models considered, we include year and country fixed effects in all specifications. In the second one, we omit fixed effects, so as to exploit between-country variation in the estimates of our effects, and in the third one, we explicitly account for the particular nature of specifications including interaction terms, following Chipman

(1996) and Crespo-Cuaresma (2011). In order to ensure the interpretability of the parameter estimates in models including interaction terms, Chipman (1996) proposes to assign a lower prior probability to specifications that contain the interacted variables without including the covariates that form it (the parent variables). We use a strong heredity prior that assigns a zero prior model probability to those specifications that include the interaction, but where the parent variables are not included as additional covariates in the model.⁸

3. Empirical results: What drives synchronization in government bond yields?

3.1. BMA results

Table 1 presents the results of the BMA exercise based on specifications of the type given by Eq. (4). For all three different settings (including country and year fixed effects, omitting them, and including heredity priors), we use the first 10,000 draws of the Markov Chain as burn-ins and base our inference on 10,000,000 draws.⁹ The selected hyperparameter on Zellner's g -prior for the regression coefficients is based on the BRIC prior, and the prior over the model space corresponds to the beta-binomial prior on inclusion probabilities by Ley and Steel (2009b), elicited so as to imply an uninformative distribution a priori over the size of the model. We include three posterior statistics of interest in Table 1: the posterior inclusion probability (PIP) of each covariate, which summarizes its relevance by presenting the sum of posterior inclusion probabilities of specifications including that particular variable, and the corresponding mean and standard deviation of the posterior distribution of the parameter attached to the variable.

The results of the BMA exercise reveal the importance of persistence dynamics to explain sovereign bond yield synchronization. For all prior settings, only autoregressive terms of the synchronization measure achieve high PIPs, and depending on the prior setting, robust differences in persistence and in the long-run mean of the synchronization variable are identified in different periods, as identified by the emergence of the zero lower bound and Mario Draghi's speech. The results in settings where no heredity prior is employed show that in recessions and periods for which the Euribor was below zero, as well as in the post-Draghi period, robust changes in the partial correlation between lagged values of the synchronization variable and current ones are observed. On the other hand, the setting which imposes a strong heredity prior only indicates different dynamics for the synchronization variable in the zero lower bound and post-Draghi periods as compared to the rest of the sample.

Concentrating on the results based on a strong heredity prior, which is a more conservative setting in terms of finding robust structural breaks, the results indicate a different level and volatility of the synchronization indicator in periods that are characterized by the zero lower bound of interest rates being a binding constraint. The estimated coefficients indicate dynamics of the synchronization measure that tend to be less volatile and less persistent during the zero lower bound period as compared to the rest of the sample. This result is in line with the view that the ECB's unconventional monetary policy was able to increase the connectedness of long-term interest rate returns across countries of the euro area (see for example Akovali and Yilmaz, 2021; Malliaropoulos and Migiakakis, 2018).

Our results show that economic fundamentals are not able to robustly predict the synchronization level of government bond yields in

the EMU. This conclusion is in conflict with the results often found in the sovereign yield spreads literature, which suggests that variables describing the fiscal position of countries are strong predictors of government bond yield spreads and thus would affect the synchronization of yields within the monetary union. The results of our model averaging exercise, on the other hand, show that modeling structural breaks in the dynamics of synchronization for particular periods related to the economic environment in which bond markets operate is capable of explaining the joint dynamics of long-term government bond yields better than specifications based on macroeconomic indicators.

The fact that macroeconomic fundamentals are found not to predict long-term yield synchronization rates relates to the nature of the synchronization measure employed. As stated previously, this measure is driven by a direct comparison between the yield volatility in the union and a counterfactual union in which a particular country is not included. This means that the inclusion of a particular country in the Union might induce higher yield volatility, irrespective of whether this is due to the country experiencing higher or lower yield levels relative to other countries in the Union.

With this in mind, the nature of our synchronization exercise would call for the use of similar synchronization measures as explanatory variables in the model. By including covariates based on synchronization indices instead of levels of the potential covariates, we are able to address whether desynchronization in particular macroeconomic variables acts as robust predictors of bond yield desynchronization in EMU. For that purpose, we repeat our BMA analysis but replace the controls with synchronization indices for that particular variable, making use of Eq. (3). Using these indices as potential covariates, the results of applying BMA with the same settings as those for the case in Table 1 are presented in Table 2.¹⁰

Similarly to the results based on covariates in levels, we find that government bond yield synchronization is a persistent process whose dynamics changed in the zero lower bound and the post-Draghi periods and appear different in the group of GIIPS countries. In addition, the results suggest that inflation synchronization indices are robust predictors of long-term government bond yields and that such an effect also differs quantitatively for the zero lower bound period and GIIPS countries. The overall effect of inflation differentials, as measured by the sum of the coefficients of the lagged inflation synchronization variables, appears larger in these subsamples, indicating a larger sensitivity of government bond yield synchronization to inflation synchronization. This result indicates that monitoring inflation differentials can be of particular importance as a leading indicator of asymmetric shocks to government bond yields in the euro area.

3.2. Exploring the jointness of yield synchronization determinants

In order to explore the nature of the models that perform particularly well in terms of explaining the synchronization of yields in EMU, we analyze the posterior results of the BMA analysis by assessing the joint posterior inclusion probability of pairs of variables. In particular, such an analysis can help us assess the importance of explicitly modeling parameter heterogeneity through the inclusion of interaction terms in specifications aimed at explaining yield synchronization. For this purpose, we compute jointness measures that quantify the likelihood that pairs of variables tend to appear together in specifications with high posterior model probability. We employ jointness measures of

⁸ See also Papageorgiou (2011) and Moser and Hofmarcher (2014) for a discussion on prior distributions over the model space for specifications with related regressors.

⁹ Standard checks for convergence of the Markov chain, such as the correlation between the relative frequency of models sampled and their analytical posterior model probability, indicate convergence.

¹⁰ For the sake of completeness, we also conducted a similar analysis for a setting where the government bond yield spreads (and not the synchronization measures) are used as the dependent variable and the potential independent variables enter the model in levels. The results are presented in the appendix and confirm the conclusion that public debt and inflation dynamics are robust drivers of yield differentials (especially when interacted with the GIIPS variable), although they do not appear to predict differences in synchronization measures based on government bond yields.

Table 2

Bayesian model averaging results (covariates as synchronization indices).

	Time and country fixed effects			Standard BMA			Strong heredity prior		
	PIP	Post. Mean	Post. SD	PIP	Post. Mean	Post. SD	PIP	Post. Mean	Post. SD
$synch_{t-1}$	1	0,7685	0,0539	1	0,8029	0,0669	1	1,1141	0,051
$synch_{t-2}$	0,0173	-0,0034	0,0278	0,0286	-0,0059	0,0376	1	-0,2705	0,0713
$synch_{t-3}$	0,0683	-0,0105	0,0411	0,0713	-0,0105	0,0401	0,994	0,0084	0,1499
$synch_{t-4}$	0,3764	0,0518	0,0728	0,6654	0,0947	0,0753	0,4416	0,1021	0,119
$uncert_{t-1}$	0,0015	0	0,0003	0,0015	0	0,0003	0,001	0	0,0002
$uncert_{t-2}$	0,0014	0	0,0003	0,0021	0	0,0004	0,0019	0	0,0005
$uncert_{t-3}$	0,0016	0	0,0004	0,0019	0	0,0004	0,0018	0	0,0004
$uncert_{t-4}$	0,0011	0	0,0003	0,0012	0	0,0003	0,001	0	0,0002
gdp_{t-1}	0,0012	0	0,0004	0,0018	0	0,0004	0,001	0	0,0003
gdp_{t-2}	0,0011	0	0,0004	0,0014	0	0,0003	0,0009	0	0,0003
gdp_{t-3}	0,0081	0,0002	0,002	0,0051	0,0001	0,0011	0,0031	0	0,0013
gdp_{t-4}	0,0015	0	0,0004	0,0013	0	0,0003	0,0023	0	0,0006
bop_{t-1}	0,0013	0	0,0003	0,0059	0,0001	0,0012	0,0019	0	0,0005
bop_{t-2}	0,0013	0	0,0004	0,002	0	0,0004	0,0013	0	0,0003
bop_{t-3}	0,002	0	0,0006	0,0015	0	0,0003	0,0012	0	0,0003
bop_{t-4}	0,001	0	0,0003	0,0028	0	0,0006	0,0012	0	0,0003
$debt_{t-1}$	0,0055	0,0002	0,0033	0,006	0,0002	0,0033	0,2038	0,0084	0,0179
$debt_{t-2}$	0,0172	0,0016	0,0167	0,007	0,0003	0,004	0,3634	0,0168	0,0287
$debt_{t-3}$	0,0056	0,0002	0,0032	0,0053	0,0002	0,0028	0,0534	0,0007	0,0134
$debt_{t-4}$	0,0087	-0,0007	0,0129	0,0049	0,0001	0,0029	0,0368	-0,0004	0,0137
inf_{t-1}	0,0014	0	0,0004	0,0012	0	0,0004	0,0011	0	0,0005
inf_{t-2}	0,0022	0	0,0008	0,0017	0	0,0005	0,0013	0	0,0006
inf_{t-3}	0,0021	0	0,0012	0,002	0	0,0009	1	0,0498	0,0212
inf_{t-4}	0,0016	0	0,0009	0,0028	0	0,0012	1	-0,0722	0,0216
$I(rec_{it} = 1)$	0,0126	-0,0002	0,0024	0,0135	-0,0003	0,0024	0,0276	-0,0003	0,002
$I(rec_{it} = 1) \times synch_{t-1}$	0,9925	0,2653	0,0531	0,9755	0,2606	0,0622	0,0241	0,0023	0,0174
$I(rec_{it} = 1) \times synch_{t-2}$	0,9791	-0,2039	0,0516	0,9704	-0,2109	0,0555	0,0052	-0,0007	0,0102
$I(rec_{it} = 1) \times synch_{t-3}$	0,0104	0,0008	0,0107	0,0086	0,0006	0,0096	0,0005	0	0,0012
$I(rec_{it} = 1) \times synch_{t-4}$	0,0665	0,0049	0,0192	0,0558	0,0044	0,019	0,0002	0	0,0005
$I(rec_{it} = 1) \times uncert_{t-1}$	0,0014	0	0,0003	0,0018	0	0,0004	0	0	0
$I(rec_{it} = 1) \times uncert_{t-2}$	0,0025	0	0,0006	0,0033	0	0,0007	0	0	0
$I(rec_{it} = 1) \times uncert_{t-3}$	0,0012	0	0,0003	0,001	0	0,0002	0	0	0
$I(rec_{it} = 1) \times uncert_{t-4}$	0,0011	0	0,0002	0,0011	0	0,0002	0	0	0
$I(rec_{it} = 1) \times gdp_{t-1}$	0,0021	0	0,0006	0,0022	0	0,0005	0	0	0
$I(rec_{it} = 1) \times gdp_{t-2}$	0,0012	0	0,0003	0,0012	0	0,0003	0	0	0
$I(rec_{it} = 1) \times gdp_{t-3}$	0,0019	0	0,0005	0,0018	0	0,0004	0	0	0
$I(rec_{it} = 1) \times gdp_{t-4}$	0,0037	0	0,0009	0,0023	0	0,0005	0	0	0
$I(rec_{it} = 1) \times bop_{t-1}$	0,0015	0	0,0004	0,0034	0	0,0008	0	0	0
$I(rec_{it} = 1) \times bop_{t-2}$	0,0015	0	0,0004	0,0037	0	0,0008	0	0	0
$I(rec_{it} = 1) \times bop_{t-3}$	0,0013	0	0,0003	0,0015	0	0,0003	0	0	0
$I(rec_{it} = 1) \times bop_{t-4}$	0,0015	0	0,0004	0,0044	0	0,0009	0	0	0
$I(rec_{it} = 1) \times debt_{t-1}$	0,0076	0,0002	0,0024	0,0127	0,0004	0,0039	0,0001	0	0,0002
$I(rec_{it} = 1) \times debt_{t-2}$	0,0185	0,0006	0,0046	0,028	0,001	0,0067	0,0003	0	0,0006
$I(rec_{it} = 1) \times debt_{t-3}$	0,0076	0,0002	0,0025	0,0102	0,0003	0,0037	0,0001	0	0,0002
$I(rec_{it} = 1) \times debt_{t-4}$	0,0116	0,0003	0,0032	0,0172	0,0005	0,0045	0,0001	0	0,0003
$I(rec_{it} = 1) \times inf_{t-1}$	0,0024	0	0,0009	0,0042	0,0001	0,0015	0	0	0
$I(rec_{it} = 1) \times inf_{t-2}$	0,0052	0,0002	0,0026	0,0126	0,0004	0,0043	0	0	0
$I(rec_{it} = 1) \times inf_{t-3}$	0,0451	-0,0021	0,0112	0,1684	-0,0073	0,019	0,0059	-0,0003	0,0036
$I(rec_{it} = 1) \times inf_{t-4}$	0,0187	0,001	0,008	0,0436	0,0023	0,0115	0,0008	0	0,0015
$I(GIIPS_i = 1)$	-	-	-	0,0027	0	0,0006	1	-0,0049	0,0091
$I(GIIPS_i = 1) \times synch_{t-1}$	0,0218	0,0055	0,04	0,0232	0,0046	0,0334	0,0185	0,0021	0,017
$I(GIIPS_i = 1) \times synch_{t-2}$	0,0649	-0,0129	0,0558	0,1111	-0,0197	0,0612	0,0025	-0,0001	0,0086
$I(GIIPS_i = 1) \times synch_{t-3}$	0,024	-0,0003	0,0243	0,0192	-0,0014	0,0209	0,0172	0,0017	0,0148
$I(GIIPS_i = 1) \times synch_{t-4}$	0,2467	0,0367	0,0678	0,2391	0,042	0,0789	0,0071	0,0007	0,0085
$I(GIIPS_i = 1) \times uncert_{t-1}$	0,001	0	0,0002	0,0013	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-2}$	0,0013	0	0,0004	0,0018	0	0,0004	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-3}$	0,0022	0	0,0006	0,0019	0	0,0004	0	0	0
$I(GIIPS_i = 1) \times uncert_{t-4}$	0,0014	0	0,0003	0,0012	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times gdp_{t-1}$	0,0024	0	0,0006	0,0018	0	0,0005	0	0	0,0001
$I(GIIPS_i = 1) \times gdp_{t-2}$	0,0012	0	0,0004	0,0014	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times gdp_{t-3}$	0,0089	0,0002	0,0021	0,005	0,0001	0,001	0,0001	0	0,0003
$I(GIIPS_i = 1) \times gdp_{t-4}$	0,0012	0	0,0003	0,0013	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times bop_{t-1}$	0,0015	0	0,0004	0,0066	0,0001	0,0013	0	0	0
$I(GIIPS_i = 1) \times bop_{t-2}$	0,0013	0	0,0003	0,0019	0	0,0004	0	0	0
$I(GIIPS_i = 1) \times bop_{t-3}$	0,0019	0	0,0005	0,0015	0	0,0003	0	0	0
$I(GIIPS_i = 1) \times bop_{t-4}$	0,001	0	0,0003	0,0031	0	0,0007	0	0	0
$I(GIIPS_i = 1) \times debt_{t-1}$	0,0206	0,0011	0,008	0,1757	0,0112	0,0253	0,0033	0,0002	0,0037
$I(GIIPS_i = 1) \times debt_{t-2}$	0,1209	0,0135	0,0465	0,4205	0,0323	0,0485	0,0082	0,0005	0,0059
$I(GIIPS_i = 1) \times debt_{t-3}$	0,0172	0,0008	0,007	0,1189	0,007	0,0202	0,0004	0	0,0011
$I(GIIPS_i = 1) \times debt_{t-4}$	0,0449	-0,0054	0,0337	0,1041	0,0007	0,0321	0,0005	0	0,0012
$I(GIIPS_i = 1) \times inf_{t-1}$	0,0009	0	0,0004	0,0013	0	0,0005	0	0	0

(continued on next page)

Table 2 (continued).

$I(GIIPS_i = 1) \times inf_{i-2}$	0,0025	0	0,0011	0,0029	0	0,0012	0	0	0,0001
$I(GIIPS_i = 1) \times inf_{i-3}$	1	-0,1299	0,0222	1	-0,1331	0,0205	1	-0,1364	0,0188
$I(GIIPS_i = 1) \times inf_{i-4}$	1	0,1327	0,0224	1	0,1382	0,0221	1	0,1604	0,0186
$I(ZLB_i = 1)$	0,0016	0	0,0018	0,0019	0,0001	0,002	1	0,0064	0,0112
$I(ZLB_i = 1) \times synch_{i-1}$	1	-1,0912	0,0743	1	-1,0989	0,0776	1	-1,0883	0,0652
$I(ZLB_i = 1) \times synch_{i-2}$	0,9996	0,8017	0,1462	0,9999	0,7767	0,138	1	1,0481	0,0709
$I(ZLB_i = 1) \times synch_{i-3}$	0,8679	0,2049	0,1097	0,8954	0,2229	0,1027	0,0066	0,0011	0,0175
$I(ZLB_i = 1) \times synch_{i-4}$	0,0752	0,0168	0,0636	0,0797	0,0175	0,0636	0,001	0,0001	0,0041
$I(ZLB_i = 1) \times uncert_{i-1}$	0,0013	0	0,0003	0,0014	0	0,0003	0	0	0
$I(ZLB_i = 1) \times uncert_{i-2}$	0,0012	0	0,0003	0,0017	0	0,0003	0	0	0
$I(ZLB_i = 1) \times uncert_{i-3}$	0,0011	0	0,0003	0,0013	0	0,0003	0	0	0
$I(ZLB_i = 1) \times uncert_{i-4}$	0,0012	0	0,0003	0,0014	0	0,0003	0	0	0
$I(ZLB_i = 1) \times gdp_{i-1}$	0,0013	0	0,0003	0,0016	0	0,0004	0	0	0
$I(ZLB_i = 1) \times gdp_{i-2}$	0,0014	0	0,0003	0,0018	0	0,0004	0	0	0
$I(ZLB_i = 1) \times gdp_{i-3}$	0,0016	0	0,0006	0,0009	0	0,0002	0	0	0,0002
$I(ZLB_i = 1) \times gdp_{i-4}$	0,0011	0	0,0003	0,001	0	0,0002	0	0	0
$I(ZLB_i = 1) \times bop_{i-1}$	0,0011	0	0,0003	0,0014	0	0,0004	0	0	0
$I(ZLB_i = 1) \times bop_{i-2}$	0,0011	0	0,0003	0,0013	0	0,0003	0	0	0
$I(ZLB_i = 1) \times bop_{i-3}$	0,0014	0	0,0003	0,0013	0	0,0003	0	0	0
$I(ZLB_i = 1) \times bop_{i-4}$	0,0013	0	0,0003	0,0019	0	0,0004	0	0	0
$I(ZLB_i = 1) \times debt_{i-1}$	0,0053	-0,0007	0,0127	0,0046	-0,0007	0,012	0,0005	0	0,0006
$I(ZLB_i = 1) \times debt_{i-2}$	0,0039	0,0007	0,0153	0,0035	0,0006	0,0148	0,0009	0	0,0009
$I(ZLB_i = 1) \times debt_{i-3}$	0,8797	-0,5924	0,2497	0,9795	-0,6966	0,1494	0,0002	0	0,0032
$I(ZLB_i = 1) \times debt_{i-4}$	0,878	0,559	0,2396	0,9791	0,6649	0,1454	0,0001	0	0,0031
$I(ZLB_i = 1) \times inf_{i-1}$	0,0062	0,0001	0,0016	0,0041	0,0001	0,0012	0	0	0,0002
$I(ZLB_i = 1) \times inf_{i-2}$	0,0407	0,0014	0,0074	0,0164	0,0005	0,0041	0,0002	0	0,0006
$I(ZLB_i = 1) \times inf_{i-3}$	0,6088	-0,0307	0,0269	0,4636	-0,0222	0,0253	0,9969	-0,0631	0,0138
$I(ZLB_i = 1) \times inf_{i-4}$	0,0256	0,0012	0,0079	0,0463	0,0022	0,0108	0,0149	0,0008	0,0072
$I(Draghi_i = 1)$	0,0016	0	0,0016	0,0014	0	0,0015	1	-0,0027	0,0112
$I(Draghi_i = 1) \times synch_{i-1}$	1	0,8772	0,0915	1	0,8806	0,0943	1	0,8036	0,0749
$I(Draghi_i = 1) \times synch_{i-2}$	0,9785	-0,6488	0,1629	0,9735	-0,6087	0,1556	1	-0,8493	0,1009
$I(Draghi_i = 1) \times synch_{i-3}$	0,134	0,0198	0,0981	0,117	0,0157	0,0965	0,4332	0,1595	0,1881
$I(Draghi_i = 1) \times synch_{i-4}$	0,748	-0,1872	0,1374	0,9614	-0,2561	0,0983	0,4367	-0,1319	0,1534
$I(Draghi_i = 1) \times uncert_{i-1}$	0,0018	0	0,0004	0,0013	0	0,0003	0	0	0
$I(Draghi_i = 1) \times uncert_{i-2}$	0,0019	0	0,0005	0,0021	0	0,0004	0	0	0
$I(Draghi_i = 1) \times uncert_{i-3}$	0,0013	0	0,0003	0,0012	0	0,0003	0	0	0
$I(Draghi_i = 1) \times uncert_{i-4}$	0,0012	0	0,0003	0,0015	0	0,0003	0	0	0
$I(Draghi_i = 1) \times gdp_{i-1}$	0,001	0	0,0003	0,0012	0	0,0003	0	0	0
$I(Draghi_i = 1) \times gdp_{i-2}$	0,001	0	0,0003	0,0015	0	0,0004	0	0	0
$I(Draghi_i = 1) \times gdp_{i-3}$	0,0018	0	0,0009	0,0013	0	0,0003	0,0004	0	0,0009
$I(Draghi_i = 1) \times gdp_{i-4}$	0,001	0	0,0003	0,0014	0	0,0003	0	0	0
$I(Draghi_i = 1) \times bop_{i-1}$	0,0013	0	0,0003	0,0016	0	0,0005	0	0	0
$I(Draghi_i = 1) \times bop_{i-2}$	0,0014	0	0,0004	0,0013	0	0,0003	0	0	0,0001
$I(Draghi_i = 1) \times bop_{i-3}$	0,002	0	0,0005	0,0015	0	0,0003	0	0	0,0001
$I(Draghi_i = 1) \times bop_{i-4}$	0,0012	0	0,0003	0,0015	0	0,0003	0	0	0
$I(Draghi_i = 1) \times debt_{i-1}$	0,0413	-0,0103	0,053	0,0345	-0,008	0,0459	0,0011	0	0,0012
$I(Draghi_i = 1) \times debt_{i-2}$	0,7343	0,2771	0,1819	0,9342	0,3554	0,1214	0,0009	0	0,0009
$I(Draghi_i = 1) \times debt_{i-3}$	0,0018	0	0,003	0,0021	0,0001	0,0039	0,0003	0	0,0006
$I(Draghi_i = 1) \times debt_{i-4}$	0,7337	-0,2393	0,1538	0,9341	-0,3184	0,1041	0,0003	0	0,0007
$I(Draghi_i = 1) \times inf_{i-1}$	0,0034	0,0001	0,0012	0,0024	0	0,0008	0	0	0,0002
$I(Draghi_i = 1) \times inf_{i-2}$	0,0119	0,0004	0,0037	0,0033	0,0001	0,0014	0	0	0
$I(Draghi_i = 1) \times inf_{i-3}$	0,005	0	0,003	0,0058	-0,0001	0,0032	0,0055	0,0003	0,0053
$I(Draghi_i = 1) \times inf_{i-4}$	0,6239	0,0401	0,0334	0,485	0,0287	0,0317	0,9951	0,0996	0,0167

Note: The table depicts the results for specifications where the yield synchronization variable is regressed against the *synchronization levels* of the independent variables. PIP stands for Posterior Inclusion Probability, Post. Mean is the mean of the posterior distribution of the corresponding effect, and Post SD is the standard deviation of the posterior distribution.

based on the work of [Doppelhofer and Weeks \(2009\)](#), [Ley and Steel \(2009a\)](#), [Ley and Steel \(2007\)](#) and [Hofmarcher et al. \(2018\)](#).¹¹ For all of the measures considered here, positive values for the jointness measure indicate that the two specific covariates at hand act as complements in the sense that they tend to appear jointly in specifications with high posterior model probability. Larger values of the jointness variables indicate that the posterior model probabilities of models which include (and/or exclude) these two covariates are larger than those which only include one of the variables (see [Amini and Parmeter, 2020](#), for a comprehensive review of jointness measures).

We start by analyzing the jointness between covariates for the space of models in which the bond yield synchronization indicators are explained by the levels of the independent variables and we include

¹¹ See the appendix for a detailed description of the measures employed, as well as the visualization of all of the jointness measures employed.

country and period fixed effects. In Figure A.1 in the online appendix, we show a heatmap of the jointness measure proposed by [Hofmarcher et al. \(2018\)](#) for this case, while the corresponding heatmaps for the other jointness indicators can be found in the appendix. The covariate pairs with the highest level of jointness include the variables that are found to have the highest individual posterior inclusion probabilities: lags of the yield synchronization variable and their interactions with the post-Draghi and zero lower bound dummies. These results point to the need of addressing structural breaks in the specification of the dynamics of yield synchronization in the period under study.¹²

¹² For the measure proposed in [Doppelhofer and Weeks \(2009\)](#), some lags of the synchronization variable interacted with the post-Draghi and recession dummies appear as strong complements with lags of the balance of payments, the zero lower bound dummy, and certain lags of GDP growth interacted with the recession variable.

We also compute jointness measures for the space of models in which we regress the bond yield synchronization variables on the synchronization indices of the independent variables. The results are presented in Figure A.2 in the online appendix for the jointness measure by Hofmarcher et al. (2018) and in the appendix for the rest of the jointness indicators. As in the previous case, the measures by Ley and Steel (2007) and Hofmarcher et al. (2018) indicate the same group of variables as those with the highest pairwise jointness. The covariates that have high jointness values are similar to those found for the case where the levels of the variables are used as covariates, expanded with inflation synchronization and its interactions with the PIIGS dummy.¹³

4. Forecasting sovereign yield synchronization rates

In this section, we examine the extent to which assessing model uncertainty can improve the forecasting accuracy of long-term yield synchronization by performing a comprehensive comparison of predictive quality across a battery of models in a pseudo out-of-sample forecasting exercise. We compare the predictive ability of model-averaged forecasts with those obtained from a panel autoregressive model (pooled AR(1), including country-fixed effects and year effects), as well as country-specific autoregressive specifications. In addition, we also evaluate the predictive accuracy of so-called median probability models, that is, single specifications that include those covariates whose posterior inclusion probability is above 0.5. The predictive ability of such a model specification is known to be optimal for selection among normal linear models (Barbieri et al., 2004).

Point forecasts of the synchronization measure obtained using BMA are based on weighting model-specific out-of-sample predictions using their corresponding posterior model probability,

$$\widehat{synch}_{it+1}^{BMA} = \sum_{j=1}^J E(synch_{it+1} | y, M_j, \{y_{it}\}_{t=1}^T) P(M_j | y, \{y_{it}\}_{t=1}^T), \quad (10)$$

where conditioning on $\{y_{it}\}_{t=1}^T$ implies that only information ranging from period 1 to period T is employed to compute the corresponding quantities.

We start by using the sample corresponding to 2001Q1–2018Q4 as an initial period sample and create one-quarter ahead predictions for all countries in the period 2019Q1. The estimation sample is then extended with the observations for 2019Q1 and the exercise is repeated recursively until the last observation of our full sample, 2020:Q4, is reached. We entertain five different types of specifications: Pooled AR(1) model, country-specific AR(1) models, BMA based on explanatory variables in levels, BMA based on explanatory variables in the form of synchronization measures, median probability model based on BMA with explanatory variables in levels and median probability model based on BMA with explanatory variables in synchronization form. We analyze the forecasting ability of each model using standard measures such as the root mean square error (RMSE), given by

$$RMSE = \sqrt{\frac{1}{C(T - T_0)} \sum_{i=1}^N \sum_{t=T_0}^T (synch_{it} - \widehat{synch}_{it})^2}, \quad (11)$$

where C is the number of countries for which forecasts are obtained, T_0 is the first period of the out-of-sample subsample and T is the final period for which data are available. In the same vein, we employ

¹³ When computing the measure by Doppelhofer and Weeks (2009), we find a substantially higher number of covariates that are strong complements. These include synchronization in inflation, uncertainty, the balance of payments, and GDP, both as parent variables and interacted with the PIIGS and zero lower bound dummies.

additionally the Theil's U statistic (Theil, 1958), which serves as an additional accuracy measure and is given by¹⁴

$$U_1 = \frac{\sqrt{\sum_{i=1}^N \sum_{t=T_0}^T (synch_{it} - \widehat{synch}_{it})^2}}{\sqrt{\sum_{i=1}^N \sum_{t=T_0}^T (synch_{it})^2}}. \quad (12)$$

We also evaluate measures that provide information on the directional accuracy of the forecasts produced by the models considered (see for example Boeck et al., 2023, for a use of these statistics in the context of prediction of credit default swap volatility). The overall directional accuracy measure is defined as the ratio of correctly predicted direction change (increase versus decrease) to all predicted changes. We complement this statistic with information about the relative number of correct predictions of increases in the synchronization measure (the hit rate) and of incorrect ones (the false alarm rate). The Kuipers Score¹⁵ is given by the difference between these two measures and is often used to evaluate forecasts of binary variables. The Hanssen and Kuipers Score is bounded between -1 and 1 , with positive values of the score implying a hit rate that is higher than the false alarm.

The results of the forecasting exercise are presented in Table 3, which shows the RMSE, Theil's U statistic, hit rate, false alarm rate, and Hanssen and Kuipers Score for the full sample of EMU countries, as well as for GIIPS and non-GIIPS economies separately.¹⁶ The predictions from country-specific AR(1) models yield the lowest RMSE systematically across all samples considered, with the second-best forecasts being achieved by the median probability model based on covariates in levels. The results for measures of out-of-sample directional accuracy indicate a very low degree of predictability in government bond yield synchronization changes, and thus a limited informational contribution of differences in macroeconomic fundamentals. A (small) positive value of the Hanssen and Kuipers Score is only achieved for the subsample of non-GIIPS countries when using BMA forecasts with both levels and synchronization covariates, and the rest of the forecasts are characterized by relatively high false alarm rates coupled with hit rates that do not tend to exceed 0.5. These findings reveal that monitoring those macroeconomic fundamentals with the highest PIPs (both in levels and synchronization dynamics) may allow policymakers, in particular those belonging to GIIPS countries, to improve predictions of short-term changes in their yield synchronization rates. For the other subsamples, the overall performance of these models and specification-averaged predictions is relatively poor.

5. Conclusions

In this paper, we examine the predictability of synchronization measures for sovereign yields within the EMU. Our findings reveal disparities in the dynamics of synchronization measures between recessions and expansion periods and distinct autocorrelation patterns during the years characterized by zero interest rates and the period following Draghi's *whatever it takes* speech. Our analysis indicates that, in general, fundamental macroeconomic variables are not able to predict government bond yield synchronization levels. The latter is attributable to our specific choice of the synchronization measure, which takes into account the structure of the monetary union. Our findings reveal that

¹⁴ Similarly to the RMSE, the Theil's U statistic relates the forecast values to the realized ones, and hence those models generating relatively lower values are preferred.

¹⁵ See Peirce (1884) for a first introduction of this measure, and Granger and Pesaran (2000) for a review of such directional accuracy measures.

¹⁶ We also present in the online appendix (Table A.5) the forecasting results when excluding the year 2020 to avoid the period which is dominated by the COVID-19 pandemic. The results obtained with an out-of-sample period ranging until 2019Q are not qualitatively different from those obtained with data including 2020.

Table 3
Results of the forecasting exercise.

	RMSE	Theil's U	DA	HR	FA	KS
All countries						
Pooled AR(1)	0.0330	0.8088	0.3571	0.4737	0.7167	−0.2430
Country-specific AR(1)	0.0089	0.2177	0.4490	0.3158	0.4667	−0.1509
BMA (levels)	0.0105	0.2563	0.3673	0.4211	0.6667	−0.2456
BMA (synchronization)	0.0153	0.3746	0.4388	0.3684	0.5167	−0.1482
Median prob. model (levels)	0.0103	0.2513	0.5000	0.4211	0.4500	−0.0289
Median prob. model (synchronization)	0.0152	0.3716	0.5000	0.5000	0.5000	0.0000
GIIPS						
Pooled AR(1)	0.0332	0.4947	0.3429	0.4706	0.7778	−0.3072
Country-specific AR(1)	0.0146	0.2179	0.3714	0.3529	0.6111	−0.2582
BMA (levels)	0.0169	0.2510	0.4286	0.3200	0.6111	−0.2911
BMA (synchronization)	0.0246	0.3668	0.4857	0.5294	0.5556	−0.0261
Median prob. model (levels)	0.0168	0.2497	0.5143	0.5294	0.5000	0.0294
Median prob. model (synchronization)	0.0244	0.3630	0.5143	0.5882	0.5556	0.0327
Non-GIIPS						
Pooled AR(1)	0.0329	3.1516	0.3651	0.4762	0.6905	−0.2143
Country-specific AR(1)	0.0022	0.2137	0.4921	0.2857	0.4048	−0.1190
BMA (levels)	0.0037	0.3552	0.3333	0.2759	0.6905	−0.4146
BMA (synchronization)	0.0054	0.5221	0.4127	0.2381	0.5000	−0.2619
Median prob. model (levels)	0.0030	0.2856	0.4921	0.3333	0.4286	−0.0952
Median prob. model (synchronization)	0.0055	0.5302	0.4921	0.4286	0.4762	−0.0476

Note: Predictive accuracy statistics based on pseudo-out-of-sample forecasts for the period 2019Q1–2020Q4, computed using a recursively expanding sample starting in 2001Q1. See text for details on the predictive accuracy measures. DA: directional accuracy; HR: hit rate; FA: false alarm rate; KS: Kuiper Score.

a rigorous assessment of model uncertainty appears necessary to assess empirically the drivers of sovereign yield misalignment within the EMU and that the benefits that can be reaped from such modeling efforts in terms of predictive improvements may be relevant for GIIPS countries, although not for other EMU members.

Our results stand in contrast with those found in the empirical literature on sovereign yield spreads, which tend to indicate that economic fundamentals (and in particular, fiscal variables) are good predictors of government bond yield spreads. Our analysis shows that, while fiscal variables may have an effect on government bond yields, they are not able to robustly explain how their synchronization across EMU countries has evolved over the last decades.

While our study has focused on the extent to which fiscal variables can help us to predict whether a country is more or less synchronized with regard to their long-term yield vis-à-vis the economic union, examining the side of financial risk is an idea worth pursuing. Further research avenues could explore how sovereign financial risk – denoted by e.g. sovereign Credit Default Swaps (CDS) – maps into sovereign yields synchronization within the EMU.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and codes to replicate the analysis can be found at <https://data.mendeley.com/datasets/r3pwwjdfvh/1>.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econmod.2024.106684>.

References

- Ahir, H., Bloom, N., Furceri, D., 2022. The World Uncertainty Index. Working Paper No. 29763, National Bureau of Economic Research, <http://dx.doi.org/10.3386/w29763>.
- Akova, U., Yilmaz, K., 2021. Unconventional Monetary Policy and Bond Market Connectedness in the New Normal. Working Paper No. 2101, Koç University-TÜSIAD Economic Research Forum (ERF), Istanbul, Retrieved from <http://hdl.handle.net/10419/243004>.
- Amini, S., Parmeter, C.F., 2020. A review of the “BMS” package for R with focus on jointness. *Econometrics* 8 (1), 6. <http://dx.doi.org/10.3390/econometrics8010006>.
- Arčabić, V., Škrinjarić, T., 2021. Sharing is caring: Spillovers and synchronization of business cycles in the European Union. *Econ. Model.* 96, 25–39.
- Arghyrou, M.G., Kontonikas, A., 2012. The EMU sovereign-debt crisis: Fundamentals, expectations and contagion. *J. Int. Final. Mark. Inst. Money* 22 (4), 658–677. <http://dx.doi.org/10.1016/j.intfin.2012.03.003>.
- Arghyrou, M.G., Tsoukalas, J.D., 2011. The Greek debt crisis: Likely causes, mechanics and outcomes. *World Econ.* 34 (2), 173–191. <http://dx.doi.org/10.1111/j.1467-9701.2011.01328.x>.
- Aßmann, C., Boysen-Hogrefe, J., 2011. Determinants of government bond spreads in the Euro Area: in good times as in bad. *Empirica* 39, <http://dx.doi.org/10.1007/s10663-011-9171-6>.
- Attinasi, M.G., Checherita-Westphal, C., Nickel, C., 2009. What Explains the Surge in Euro Area Sovereign Spreads During the Financial Crisis of 2007–09?. Working Paper Series No. 1131, European Central Bank, <http://dx.doi.org/10.2139/ssrn.1520351>.
- Barbieri, M.M., Berger, J.O., Tre, U.R., 2004. Optimal predictive model selection. *Ann. Statist.* 32 (3), 870–897. <http://dx.doi.org/10.1214/009053604000000238>.
- Barrios, S., Iversen, P., Lewandowska, M., Setzer, R., 2009. Determinants of Intra-Euro Area Government Bond Spreads During the Financial Crisis. *European Economy - Economic Papers* 2008 - 2015 No. 388, Directorate General Economic and Financial Affairs (DG ECFIN), European Commission, <http://dx.doi.org/10.2765/29413>.
- Bernoth, K., Von Hagen, J., Schuknecht, L., 2012. Sovereign risk premiums in the European government bond market. *J. Int. Money Finance* 31 (5), 975–995. <http://dx.doi.org/10.1016/j.jimonfin.2011.12.006>.
- Boeck, M., Feldkircher, M., Raunig, B., 2023. A view from outside: sovereign CDS volatility as an indicator of economic uncertainty. *Macroecon. Dyn.* 1–28. <http://dx.doi.org/10.1017/S1365100523000524>.
- Burnside, C., Eichenbaum, M., Rebelo, S., 2016. Currency crises models. In: *Banking Crises: Perspectives from the New Palgrave Dictionary*. Springer, pp. 79–83.
- Chipman, H., 1996. Bayesian variable selection with related predictors. *Canad. J. Statist.* 24 (1), 17–36. <http://dx.doi.org/10.2307/3315687>.
- Clayey, P., Vasicek, B., 2012. Measuring Sovereign Bond Spillover in Europe and the Impact of Rating News. AQR-Working Papers, 2012, AQR12/09.

- Claeys, P., Vašíček, B., 2014. Measuring bilateral spillover and testing contagion on sovereign bond markets in Europe. *J. Bank. Financ.* 46, 151–165. <http://dx.doi.org/10.1016/j.jbankfin.2014.05.011>.
- Costantini, M., Fragetta, M., Melina, G., 2014. Determinants of sovereign bond yield spreads in the EMU: An optimal currency area perspective. *Eur. Econ. Rev.* 70, 337–349. <http://dx.doi.org/10.1016/j.euroecorev.2014.06.004>.
- Crespo-Cuaresma, J., 2011. How different is Africa? A comment on Masanjala and Papageorgiou. *J. Appl. Econometrics* 26 (6), 1041–1047. <http://dx.doi.org/10.1002/jae.1180>.
- Crespo-Cuaresma, J., Fernández-Amador, O., 2013a. Business cycle convergence in EMU: A first look at the second moment. *J. Macroecon.* 37, 265–284. <http://dx.doi.org/10.1016/j.jmacro.2013.02.001>.
- Crespo-Cuaresma, J., Fernández-Amador, O., 2013b. Business cycle convergence in EMU: A second look at the second moment. *J. Int. Money Finance* 37, 239–259. <http://dx.doi.org/10.1016/j.jimonfin.2013.06.002>.
- De Grauwe, P., Ji, Y., 2013. Self-fulfilling crises in the Eurozone: An empirical test. *J. Int. Money Finance* 34, 15–36. <http://dx.doi.org/10.1016/j.jimonfin.2012.11.003>.
- Doppelhofer, G., Weeks, M., 2009. Jointness of growth determinants. *J. Appl. Econometrics* 24 (2), 209–244. <http://dx.doi.org/10.1002/jae.1046>.
- Engen, E.M., Hubbard, R.G., 2004. Federal government debt and interest rates. *NBER Macroecon. Annu.* 19, 83–138.
- Fernández, C., Ley, E., Steel, M.F.J., 2001a. Benchmark priors for Bayesian model averaging. *J. Econometrics* 100 (2), 381–427. [http://dx.doi.org/10.1016/S0304-4076\(00\)00076-2](http://dx.doi.org/10.1016/S0304-4076(00)00076-2).
- Fernández, C., Ley, E., Steel, M.F.J., 2001b. Model uncertainty in cross-country growth regressions. *J. Appl. Econometrics* 16 (5), 563–576. <http://dx.doi.org/10.1002/jae.623>.
- Gapen, M., Gray, D., Lim, C.H., Xiao, Y., 2008. Measuring and analyzing sovereign risk with contingent claims. *IMF Staff Pap.* 55 (1), 109–148. <http://dx.doi.org/10.1057/palgrave.imfsp.9450026>.
- Granger, C.W., Pesaran, M.H., 2000. Economic and statistical measures of forecast accuracy. *J. Forecast.* 19 (7), 537–560. [http://dx.doi.org/10.1002/1099-131X\(200012\)19:7%3C537::AID-FOR769%3E3.0.CO;2-G](http://dx.doi.org/10.1002/1099-131X(200012)19:7%3C537::AID-FOR769%3E3.0.CO;2-G).
- Haugh, D., Ollivaud, P., Turner, D., 2009. What Drives Sovereign Risk Premiums?: An Analysis of Recent Evidence from the Euro Area. OECD Economics Department Working Papers No. 718, <http://dx.doi.org/10.1787/222675756166>.
- Hofmarcher, P., Cuaresma, J.C., Grün, B., Humer, S., Moser, M., 2018. Bivariate jointness measures in Bayesian model averaging: Solving the conundrum. *J. Macroecon.* 57, 150–165. <http://dx.doi.org/10.1016/j.jmacro.2018.05.005>.
- Hwang, S.H., Kim, Y.J., 2021. International output synchronization at different frequencies. *Econ. Model.* 104, 105627.
- Kilponen, J., Laakkonen, H., Vilmunen, J., 2015. Sovereign risk, European crisis resolution policies and bond yields. *Int. J. Central Bank.* 11 (2), 285–323. <http://dx.doi.org/10.2139/ssrn.2102461>.
- Laubach, T., 2009. New evidence on the interest rate effects of budget deficits and debt. *J. Eur. Econ. Assoc.* 7 (4), 858–885. <http://dx.doi.org/10.1162/JEEA.2009.7.4.858>.
- Ley, E., Steel, M.F.J., 2007. Jointness in Bayesian variable selection with applications to growth regression. *J. Macroecon.* 29 (3), 476–493. <http://dx.doi.org/10.1016/j.jmacro.2006.12.002>.
- Ley, E., Steel, M.F.J., 2009a. Comments on ‘Jointness of growth determinants’. *J. Appl. Econometrics* 24 (2), 248–251. <http://dx.doi.org/10.1002/jae.1049>.
- Ley, E., Steel, M.F.J., 2009b. On the effect of prior assumptions in Bayesian model averaging with applications to growth regression. *J. Appl. Econ.* 24 (4), 651–674. <http://dx.doi.org/10.1002/jae.1057>.
- Malliaropoulos, D., Migiakis, P., 2018. Quantitative easing and sovereign bond yields: a global perspective. In: *Working Papers of the Bank of Greece*.
- Manasse, P., Schimmlerpfennig, M.A., Roubini, N., 2003. Predicting Sovereign Debt Crises. International Monetary Fund, <http://dx.doi.org/10.5089/9781451875256.001>.
- Martins, J., 2022. Bond yields movement similarities and synchronization in the G7: A time–frequency analysis. *J. Bus. Cycle Res.* 18 (2), 189–214. <http://dx.doi.org/10.2139/ssrn.872705>.
- Moser, M., Hofmarcher, P., 2014. Model priors revisited: Interaction terms in bma growth applications. *J. Appl. Econometrics* 29 (2), 344–347. <http://dx.doi.org/10.1002/jae.2365>.
- Obstfeld, M., 1986. Speculative attack and the external constraint in a maximizing model of the balance of payments. *Can. J. Econ. / Rev. Can. D'econom.* 19 (1), 1–22. <http://dx.doi.org/10.2307/135168>.
- Obstfeld, M., 1995. The logic of currency crises. In: *Monetary and Fiscal Policy in an Integrated Europe*. Springer, pp. 62–90.
- Papageorgiou, C., 2011. How to use interaction terms in BMA: Reply to Crespo Cuaresma's comment on Masanjala and Papageorgiou (2008). *J. Appl. Econometrics* 26 (6), 1048–1050. <http://dx.doi.org/10.1002/jae.1263>.
- Peirce, C.S., 1884. The numerical measure of the success of predictions. *Science* (93), 453–454. <http://dx.doi.org/10.1126/science.ns-4.93.453.b>.
- Poghosyan, T., 2014. Long-run and short-run determinants of sovereign bond yields in advanced economies. *Econ. Syst.* 38 (1), 100–114. <http://dx.doi.org/10.1016/j.ecosys.2013.07.008>.
- Theil, H., 1958. *Economist Forecasts and Policy*. North-Holland Publishing Company, Amsterdam-London, <http://dx.doi.org/10.2307/2229320>.
- Zellner, A., 1986. On assessing prior distributions and Bayesian regression analysis with g-prior distributions. In: *Bayesian Inference and Decision Techniques*. Elsevier Science.